# Preliminary Project Plan

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## 1. Project Overview

The goal of this project is to develop a Python-based structural analysis program using the Direct Stiffness Method (DSM) for mixed three-dimensional truss and frame systems. The solver will allow different element types to be assigned member by member within the same structural model, providing a unified framework for mixed structural analysis.

The structural model will be defined through an external input file so that the user can modify the problem without editing the source code directly. Based on this input, the program will perform preprocessing, assemble the global structural system, solve for the unknown response quantities, and generate organized output files together with selected visualization results.

In addition to computing structural response, the project also aims to present the results clearly and intuitively. Visualization of structural behavior will help identify important members, critical regions, and response trends. To verify the correctness of the implementation, separate validation examples for truss and frame behavior will be used. After validation, the developed code will be applied to a mixed structural system to establish a complete analysis workflow.

## 2. Code Organization

The GitHub repository for this project is available at the following link:

`https://github.com/kimchi-chisung/CEE6501_MSA_final_course_project`

The repository will be organized into separate folders for input files, output files, source code, the report, and presentation materials. Input files will be divided into validation examples and final structural analysis cases, and the outputs will also be stored separately for each case.

The code will be organized into a Jupyter notebook and separate Python module files. The overall execution workflow will be handled through `main.ipynb`, while the main calculations and supporting functions will be stored in separate Python modules.

The source code will be divided by function, including modules for input processing, degree-of-freedom handling, truss and frame element calculations, global assembly, solving, and postprocessing/visualization. Most of these modules will be reorganized from code developed in previous assignments, with additional modules added for mixed truss–frame analysis, color-based visualization, and comparative scenario studies.

## 3. Project Components

This project is expected to consist of four main components. The first component is user input processing. The user will define the structural model through an external input file containing nodal coordinates, element connectivity, element type, material and section properties, support conditions, nodal loads, member loads, and prescribed displacements. The program will read this file and preprocess it into internal data structures required for analysis.

The second component is the main analysis module. In this part, the global stiffness matrix and load vector will be assembled using the Direct Stiffness Method, and the unknown displacements and reactions will be solved after applying the boundary conditions. This module will handle both three-dimensional truss elements and three-dimensional frame elements within the same codebase, and will include required features such as frame member loads, release conditions, temperature loading, fabrication error, and prescribed support displacements. It will also allow the same structure to be analyzed under different loading, support, and member-condition scenarios.

The third component is validation examples. In order to confirm that the program works correctly, I plan to compare the code results with hand calculations using simple truss and frame examples. If necessary, validation may also be carried out by embedding two-dimensional problems within the three-dimensional code framework.

The fourth component is the final structural modeling study and result presentation. In the final stage, I plan to analyze a realistic structure that contains both truss and frame elements. This will include structural idealization, support and loading assumptions, interpretation of the results, and comparative studies under selected scenario changes. The results will be presented not only as numerical output but also through tables, text files, and visualization figures.

## 4. Proposed Final Structure and Preliminary Background Review

The final structure selected for this project is the Chaotianmen Bridge in Chongqing, China. This bridge is a continuous steel truss arch bridge with tie girders crossing the Yangtze River, with a main span of 552 m and a total length of 1,741 m. It is also a double-deck bridge carrying both roadway and railway traffic.

This structure was selected because it is well suited for the combined use of frame and truss elements. The main chord members of the arch rib, the tie girder, the floorbeams, the stringers, and the major beam members of the deck system can be modeled using frame elements. In contrast, the internal web members of the arch, diagonal bracing members, and some auxiliary connecting members can be idealized as truss elements. This makes the bridge a suitable example of mixed three-dimensional truss–frame analysis.

The preliminary background review focused on three points. First, I examined how similar arch

Figure 1: Chaotianmen Bridge in Chongqing, China.

bridges and steel truss arch bridges have been modeled in previous studies. Song et al. analyzed a continuous steel truss arch railway bridge using a three-dimensional numerical model combined with monitoring data [1]. Second, I reviewed how loading conditions, support conditions, and temperature effects have been considered in arch bridge studies. Sun et al. analyzed an arch bridge system including steel truss beams, suspenders, stringers, and lateral beams [2]. Third, I reviewed how local member damage can affect global structural behavior. Fan et al. investigated the influence of hanger damage and arch-beam combination parameters on a through tied-arch bridge [3].

Based on this review, several what-if scenarios will be considered in the final study, including temperature change, support-condition change, asymmetric live loading, and local stiffness reduction or removal of selected truss members or hangers. Overall, the Chaotianmen Bridge is an appropriate final modeling case because of its span scale, structural form, and member configuration that naturally accommodates both truss and frame idealization.

## References

[1] Y. sheng Song, Y. liang Ding, H. wei Zhao, G. xin Wang, A. qun Li, and B. ya Cao, "Structural behavior analysis of a continuous steel truss arch railway bridge integrating monitoring data," *Advances in Mechanical Engineering*, vol. 9, no. 1, pp. 1–14, 2017.

[2] N. Sun, X. Zheng, Y. Li, Y. Zhao, H. Yuan, and M. Zhou, "Numerical study on the mechanical performance of a flexible arch composite bridge with steel truss beams over its entire lifespan," *Sustainability*, vol. 16, no. 14, p. 6041, 2024.

[3] B.-H. Fan, Q. Sun, S.-G. Wang, and Q. Chen, "Numerical analysis of failure mechanism in through tied-arch bridges: Impact of hanger damage and arch-beam combination parameters," *Symmetry*, vol. 17, no. 11, p. 1823, 2025.